

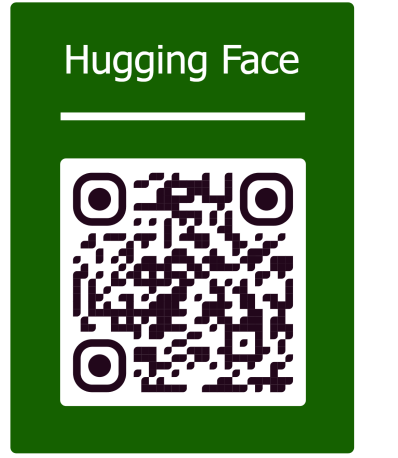
PhysicsEval: Inference-Time Techniques to Improve the Reasoning Proficiency of Large Language Models on Physics Problems



Oshayer Siddique*, J. M Areeb Uzair Alam*, Md Jobayer Rahman Rafy*,
Syed Rifat Raiyan*†, Hasan Mahmud, Md Kamrul Hasan

Department of Computer Science and Engineering
Islamic University of Technology, Dhaka, Bangladesh

{oshayer, areebuzair, jobayerrahman, rifatraiyan, hasan, hasank}@iut-dhaka.edu



Introduction

Early language models relied on quick, intuitive reasoning, which limited their ability to solve complex problems. Techniques such as Chain-of-Thought [1] and Tree-of-Thought [2] improved logical structure, yet physics remains a challenge, with models often making calculation, interpretation, and unit errors. Stronger prompting strategies and self-correcting or agent-based systems can help catch these mistakes. Our approach includes self-refinement, where the model revisits and improves its own answers, and multi-agent review, where smaller models provide verification and feedback. Using open-source verifiers reduces both computational cost and bias. To assess these methods, we introduce PHYSICSEVAL, a dataset of 19,609 physics problems, with 10% reserved for testing and the remainder for training.

Dataset Construction

We build a dataset of 19,609 physics problems collected from 20 textbooks and trusted educational websites. It covers 19 topics, like *Mechanics*, *Thermodynamics*, *Electromagnetism*, *Waves*, *Optics*, *Relativity*, and *Quantum Physics*.

The solutions, often containing mathematical or formatting errors, are passed through Gemini 2.5 Pro for corrections. The solutions are elaborated and split into steps. Further metadata, such as *soft labels*, *difficulty* etc. are added.

The data is split 90:10, giving 17,647 training problems and 1,962 test problems to help train and evaluate reasoning-focused models. The dataset is available at <https://huggingface.co/datasets/IUTVanguard/PhysicsEval>.

Methodology:

We try out four different inference-time techniques to solve Physics problems:

Baseline: A powerful LLM, designated as *Proposer*, is given a Physics problem. The solution it generates is accepted as is. This solution is the baseline for the other techniques.

Self-refine: Once the Baseline solution is generated, the Proposer is prompted to check its own answer for any mistakes before generating a new solution.

Single Agent Review: An LLM separate from the Proposer is designated as *Meta-verifier*. Meta-verifier is given the problem and the generated base solution and told to generate a list of mistakes. If any mistakes are found, the list is passed to the proposer, who generates a new solution.

Multi Agent Review: Multiple small LLMs designated as *Verifiers* are given the problem and the generated base solution and told to generate lists of mistakes. These lists are then passed onto the Meta-verifier, which generates an aggregated, logical list of mistakes. Lastly, Proposer generates a new solution based on the list.

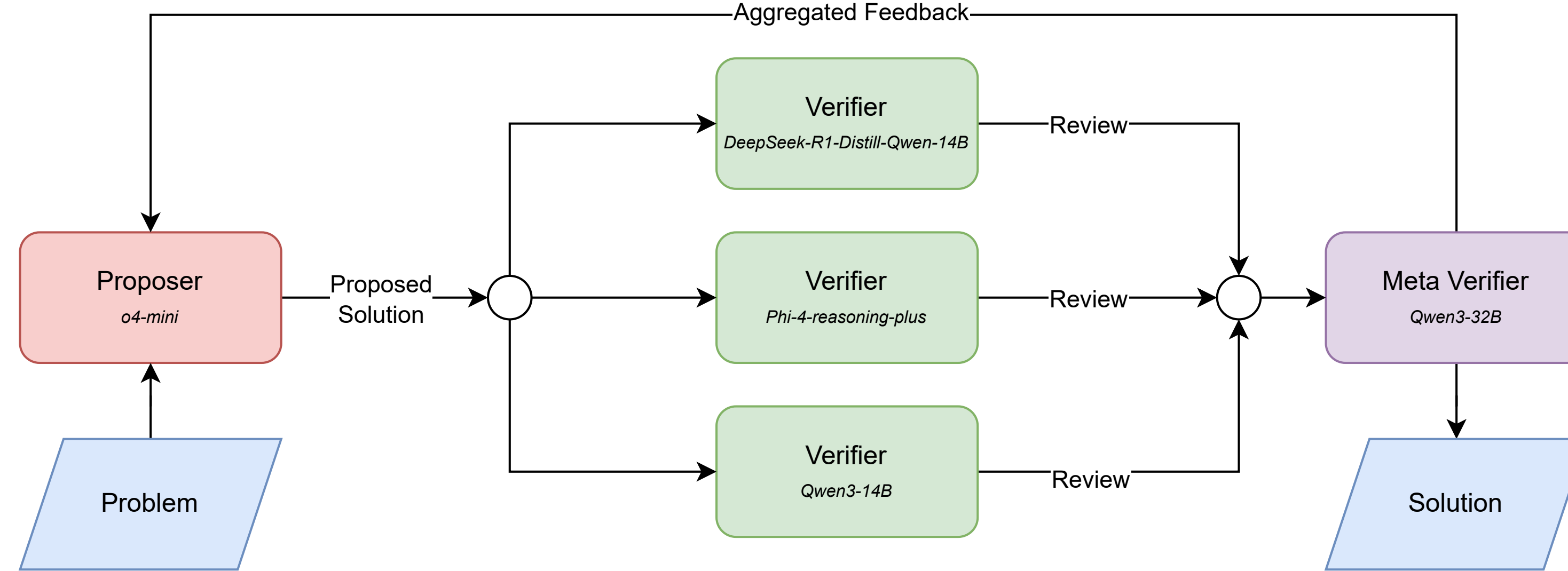


Figure 1: An overview of the multi-agent review model. The model names are, of course, subject to shuffling.

Result Analysis

We use Gemini 2.5 Pro to compare the LLM generated response to the solution present in PHYSICSEVAL. Gemini is used strictly to compare the two solutions, and a rubric of scoring is set, based on Docktor et al. (2016)[3].

Metric Component	Weight (w_k)
Mathematical Accuracy (w_{MA})	0.30
Logical Consistency (w_{LC})	0.25
Formulas and Principles (w_{FP})	0.20
Completeness (w_C)	0.10
Assumptions Made (w_A)	0.10
Clarity and Coherence (w_{CC})	0.05
Total	1.00

Upon analyzing the performances of different models, we find that self-refine causes a decrease in scores, while Single agent review and Multi agent review causes an overall increase.

Our Contributions

- We test the performance of existing LLMs in solving physics problems.
- We use multiple inference time techniques and compare their performances.
- We curate a large dataset of Physics problems containing both mathematical and descriptive problems.

Problem:
At what rate does the Sun emit photons? For simplicity, assume that the Sun's entire emission at the rate of 3.9×10^{26} W is at the single wavelength of 550 nm.

Figure 2: Example of an astrophysics problem from the PHYSICSEVAL benchmark.

Model	Easy (1-4)				Medium (5-7)				Hard (8-10)			
	Baseline	Self-Refine	Single-Agent	Multi-Agent	Baseline	Self-Refine	Single-Agent	Multi-Agent	Baseline	Self-Refine	Single-Agent	Multi-Agent
DeepSeek-R1	90.6	92.7	93.3	94.1	80.8	84.4	83.1	83.4	72.9	73.7	74.7	72.7
Gemma 3 27B	86.9	85.7	86.4	87.6	55.8	56.5	59.4	59.1	41.5	40.4	39.1	40.6
Llama 4 Maverick	91.5	90.5	92.0	92.9	83.6	82.0	82.6	82.4	55.2	54.0	57.9	52.1
Phi-4-reasoning-plus	84.4	94.4	93.2	94.7	86.9	92.2	93.4	93.9	80.1	83.4	83.8	87.6
QwQ-32B	93.7	94.0	94.2	94.6	80.9	80.8	81.1	81.9	63.9	68.6	63.5	71.0
o4-mini	86.7	82.6	85.9	86.8	87.3	86.4	87.0	88.2	83.6	82.3	84.1	85.4
Average	88.96	89.98	90.83	91.78	79.21	80.38	81.10	81.48	66.20	67.06	67.18	68.23

Table 1: Average PPS score on PHYSICSEVAL across frontier LLMs, stratified by problem difficulty and inference method. Higher scores are better. The best-performing method for each model and difficulty tier is highlighted in green, while the worst is highlighted in purple.

Comparison	Metric	Weight	t-statistic	p-value	Significant ($p < 0.05$)?
Single Agent vs. Baseline	Overall PPS	1.00	-0.01	0.996	False
Single Agent vs. Baseline	Mathematical Accuracy	0.30	0.89	0.373	False
Single Agent vs. Baseline	Logical Consistency	0.25	0.35	0.729	False
Single Agent vs. Baseline	Completeness	0.10	-2.36	0.0182	True
Single Agent vs. Baseline	Clarity And Coherence	0.05	-3.31	0.000965	True
Single Agent vs. Baseline	Formulas Principles	0.20	0.29	0.769	False
Single Agent vs. Baseline	Assumptions Made	0.10	-1.23	0.220	False
Multi-Agent Review vs. Baseline	Overall PPS	1.00	2.05	0.0405	True
Multi-Agent Review vs. Baseline	Mathematical Accuracy	0.30	2.77	0.00570	True
Multi-Agent Review vs. Baseline	Logical Consistency	0.25	1.08	0.278	False
Multi-Agent Review vs. Baseline	Completeness	0.10	-1.44	0.150	False
Multi-Agent Review vs. Baseline	Clarity And Coherence	0.05	-1.79	0.0739	False
Multi-Agent Review vs. Baseline	Formulas Principles	0.20	2.50	0.0126	True
Multi-Agent Review vs. Baseline	Assumptions Made	0.10	1.50	0.134	False

Table 2: Statistical significance of performance improvements for o4-mini. Results with a p -value < 0.05 are considered statistically significant.

Benchmark	Size	Knowledge	Question		Solution		
			Type	Avg. Tokens	Step-by-step	Avg. Tokens	Avg. Steps
JEEBench	123	CEE	OE, MC	169.7	×	—	—
MMLU-Pro	1,299	COL	MC	52.1	✓	—	—
GPQA	227	PH.D.	OE	111.4	×	197.2	3.6
SciEval	1,657	—	OE, MC	154.5	×	—	—
SciBench	295	COL	OE	80.5	×	315.9	2.8
MMMU	443	COL	OE, MC	53.8	×	—	—
ScienceQA	617	K1-K12	MC	13.3	×	63.0	2.4
OlympiadBench	2,334	COMP	OE	222.0	×	199.8	3.7
EMMA	156	—	MC	109.5	×	—	—
PhysReason	1,200	CEE+COMP	OE	226.3	✓	441.3	8.1
UGPhysics	11,040	COL	OE, MC	82.4	✓	318.5	—
PHYSICSEVAL	19,609	CEE+COL+COMP	OE	98.8	✓	3830.8	3.9

Table 3: Comparison of PHYSICSEVAL with various other physics reasoning benchmarks. For the "Knowledge" column: COMP refers to Competition level, COL to College level, CEE to College Entrance Examination, K1-K12 to elementary and high school levels, and PH.D. to Doctor of Philosophy. For the "Question Type" column: OE denotes open-ended questions, while MC stands for multiple-choice questions.

Limitations

- High computational cost: the multi-agent verification runs multiple models per problem, increasing runtime and resource use.
- Dataset not fully manually validated; many solutions were cleaned up automatically.
- Human-authored answers were stylistically refined with an LLM (Gemini 2.5 Pro); only a small sample received manual checks.
- Residual errors in the dataset are possible and could skew model training and evaluation.
- Evaluation depends heavily on LLM-based grading because ground-truth solutions are detailed and descriptive.
- Methods are optimized for physics and may require adaptation or fine-tuning to work well in other STEM domains.

Conclusion

Our evaluation of leading language models on the new PHYSICSEVAL benchmark shows clear progress in physics problem-solving, along with notable limitations. Baseline performance is solid in many areas, but our tests show that inference-time methods, especially multi-agent verification, can significantly boost accuracy, particularly on harder problems and in technical topics where initial answers are less reliable. These improvements, however, vary by category and scoring rubric, reflecting the context-dependent nature of collaborative checking. Overall, the results highlight the need for adaptive, category-specific strategies to further strengthen scientific reasoning in LLMs. We aim for PHYSICSEVAL to support continued research and development in this direction.

Useful Links

Paper: <https://arxiv.org/abs/2508.00079>

Experiment GitHub: <https://github.com/areebuzair/PhysicsEval>

Dataset: <https://huggingface.co/datasets/IUTVanguard/PhysicsEval>

References

- [1] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models, 2023.
- [2] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Tom Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. *Advances in neural information processing systems*, 36:11809–11822, 2023.
- [3] Jennifer L Docktor, Jay Dornfeld, Evan Frodermann, Kenneth Heller, Leonardo Hsu, Koblar Alan Jackson, Andrew Mason, Qing X Ryan, and Jie Yang. Assessing student written problem solutions: A problem-solving rubric with application to introductory physics. *Physical review physics education research*, 12(1):010130, 2016.